Giovanni Toto

Department of Statistics and Data Sciences, The University of Texas at Austin
2317 Speedway D9800, Austin, TX 78712-1823, USA
giovanni.toto@austin.utexas.edu

About Me

Date of birth:	May 12, 1997
Place of birth:	Padova, Italy
Nationality:	Italian

Current Position

Oct. 2025 - ongoing

Postdoctoral associate
Department of Statistics and Data Sciences, The University of Texas at Austin
Supervisor: Yang Ni

Education

Oct. 2022 - Sept. 2025

PhD in Statistics
Department of Statistical Sciences, University of Padova
Thesis title: Bayesian Nonparametric Multiple Random Partition Models
Supervisor: Antonio Canale
Co-supervisors: Peter Müller, Abhra Sarkar

Oct. 2019 - March 2022

Master's degree in Statistical Sciences
Department of Statistical Sciences, University of Padova
Thesis title: A Topic Modeling Algorithm for Microblogs
Supervisor: Emanuele Di Buccio
Final mark: 110/110 cum laude

Oct. 2016 - Nov. 2019

Bachelor's degree in Statistics for Technology and Science
Department of Statistical Sciences, University of Padova
Thesis title: Design and Implementation of a Video Search Engine and a Graphical User Interface for Sentiment Analysis
Supervisors: Massimo Melucci, Emanuele Di Buccio
Final mark: 101/110

Research Activities

June 2022 - Oct. 2022

Research fellowship
Department of Statistical Sciences, University of Padova
Title: Development of R libraries for the implementation of Bayesian models for sc-RNA-seq data
Supervisors: Antonio Canale, Davide Risso
Implementation in R and C++ of Bayesian models for sc-RNA-seq data.

March 2020 - April 2020

Research fellowship
Department of Information Engineering, University of Padova
Title: Design, implementation, and use of a computer system for online acquisition of multimodal documents and real-time opinion analysis
Supervisor: Massimo Melucci
Design and implementation of a graphical interface allowing real-time analysis of opinions expressed in a video through multimodal sentiment analysis.

Research Interests

- Bayesian nonparametrics
- Density estimation
- Functional data analysis
- Random partition models

Publications

[1] Li, Q., Stefani, A., Toto, G., Di Buccio, E., & Melucci, M. (2020). Towards multimodal sentiment analysis inspired by the quantum theoretical framework. *3rd IEEE Conference on Multimedia Information Processing and Retrieval, MIPR 2020, Shenzhen, China, August 6-8, 2020*, 177–180

[2] Toto, G., & Di Buccio, E. (2022). A modular approach to topic modeling for heterogeneous documents. In G. Pasi, P. Cremonesi, S. Orlando, M. Zanker, D. Massimo, & G. Turati (Eds.), *Proceedings of the 12th Italian Information Retrieval Workshop 2022, Milan, Italy, June 29-30, 2022*. CEUR-WS.org

[3] Canale, A., Galtarossa, L., Risso, D., Schiavon, L., & Toto, G. (2026). Structured factorization for single-cell gene expression data. *Journal of the Royal Statistical Society Series C: Applied Statistics*

Preprints

[1] Toto, G., & Canale, A. (202+). Bayesian local clustering of functional data via semi-Markovian random partitions

[2] Toto, G., Müller, P., & Sarkar, A. (202+). Bayesian semiparametric multivariate density regression with coordinate-wise predictor Selection

Conference Presentations

[1] Contributed talk: A modular approach to topic modeling for heterogeneous documents; IIR 2022, Milan, Italy (June 2022).

[2] Poster: Local Bayesian clustering for functional data; BAYSM 2024, Venice, Italy (June 2024).

[3] Poster: Local Bayesian clustering for functional data; ISBA 2024, Venice, Italy (July 2024).

[4] Poster: Bayesian multivariate density regression with coordinate-wise predictor selection; BOSS 2025, College Station, TX, USA (April 2025).

[5] Poster: Bayesian multivariate density regression with coordinate-wise predictor selection; BNP 14, Los Angeles, CA, USA (June 2025).

[6] Poster: Bayesian multivariate density regression with coordinate-wise predictor selection; SISBayes, Padova, Italy (September 2025).

[7] Poster: Bayesian multivariate density regression with coordinate-wise predictor selection; COTS, Austin, TX, USA (April 2026).

Awards

- Travel award at BNP 14, Los Angeles, CA, USA (June 2025).
- Best poster award at ISBA 2024, Venice, Italy (July 2024).